

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 08: JOURNEY INSIDE THE ATOM: STRUCTURE & COMPOSITION
TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-08

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 Who experimentally discovered the electron in 1897 using cathode ray discharge tubes?

- | | |
|------------------|-----------------------|
| (A) J.J. Thomson | (B) Ernest Rutherford |
| (C) John Dalton | (D) James Chadwick |

Q.2 Who discovered the presence of positively charged canal rays in a discharge tube in 1886?

- | | |
|------------------|------------------|
| (A) E. Goldstein | (B) J.J. Thomson |
| (C) Niels Bohr | (D) Marie Curie |

Q.3 Who discovered the uncharged subatomic particle, the neutron, in 1932?

- | | |
|--------------------|-----------------------|
| (A) James Chadwick | (B) Ernest Rutherford |
| (C) Niels Bohr | (D) John Dalton |

Q.4 Rutherford's gold foil alpha scattering experiment led to the discovery of the:

- | | |
|--------------|--------------------|
| (A) Electron | (B) Atomic Nucleus |
| (C) Neutron | (D) Quark |

Q.5 What is an alpha (α) particle?

- | | |
|---------------------|--|
| (A) A hydrogen atom | (B) A doubly charged helium ion (He^{2+}) |
| (C) A fast electron | (D) An uncharged photon |

Q.6 What is the maximum number of electrons that can be accommodated in the L-shell ($n=2$)?

- | | |
|--------|--------|
| (A) 2 | (B) 8 |
| (C) 18 | (D) 32 |

Q.7 What is the maximum number of electrons that can be held in the outermost valence shell of any atom?

- | | |
|--------|--------------------|
| (A) 2 | (B) 8 (Octet rule) |
| (C) 18 | (D) 32 |

Q.8 What is the electronic configuration of Sodium (Atomic number $Z = 11$)?

- | | |
|-------------|-------------|
| (A) 2, 8, 1 | (B) 2, 9 |
| (C) 8, 2, 1 | (D) 2, 8, 8 |

Q.9 What is the valency of Chlorine (Atomic number $Z = 17$, config 2, 8, 7)?

- | | |
|-------|-------|
| (A) 7 | (B) 1 |
| (C) 2 | (D) 8 |

Q.10 What is the valency of Argon (Atomic number $Z = 18$, config 2, 8, 8)?

- | | |
|-------|--------------|
| (A) 8 | (B) 0 (Zero) |
| (C) 2 | (D) 1 |

Q.11 The mass number (A) of an atom is defined as the total number of:

- | | |
|----------------------------|-------------------------------------|
| (A) Protons and electrons | (B) Protons and neutrons (nucleons) |
| (C) Neutrons and electrons | (D) Protons only |

Q.12 Atoms of the same element having identical atomic numbers (Z) but different mass numbers (A) are:

- | | |
|--------------|--------------|
| (A) Isobars | (B) Isotopes |
| (C) Isotones | (D) Isomers |

Q.13 Atoms of different chemical elements having different atomic numbers but identical mass numbers are:

- | | |
|----------------|-------------|
| (A) Isotopes | (B) Isobars |
| (C) Allotropes | (D) Anions |

Q.14 What is the average relative atomic mass of natural chlorine?

- | | |
|------------|------------|
| (A) 35.0 u | (B) 35.5 u |
| (C) 37.0 u | (D) 36.0 u |

Q.15 Which isotope is used as fuel in nuclear power reactors to produce electricity?

- | | |
|---------------|--------------------------------------|
| (A) Cobalt-60 | (B) Uranium-235 (^{235}U) |
| (C) Carbon-14 | (D) Iodine-131 |

Q.16 Which radioisotope is used in radiotherapy for treating cancer?

- | | |
|------------------------------------|---------------|
| (A) Cobalt-60 (^{60}Co) | (B) Carbon-14 |
| (C) Uranium-238 | (D) Argon-40 |

Q.17 Which radioisotope is used in the diagnosis and clinical treatment of thyroid goitre?

- | | |
|-------------------------------------|-------------------|
| (A) Iodine-131 (^{131}I) | (B) Cobalt-60 |
| (C) Uranium-235 | (D) Phosphorus-32 |

Q.18 Which element's most abundant neutral atom contains zero neutrons?

- | | |
|-------------|--|
| (A) Helium | (B) Hydrogen (^1_1H , Protium) |
| (C) Lithium | (D) Carbon |

Q.19 What is the mass of an electron relative to a proton?

- | | |
|------------|---|
| (A) Equal | (B) Approximately $\frac{1}{1840}$ of a proton mass |
| (C) Double | (D) Zero |

Q.20 What is the charge on a neutron?

- (A) $+1e$ (B) $-1e$
(C) Zero (electrically neutral) (D) $+2e$

Q.21 Who proposed that electrons revolve only in discrete non-radiating orbits called stationary energy levels?

- (A) Niels Bohr (B) Ernest Rutherford
(C) J.J. Thomson (D) John Dalton

Q.22 What is the valency of Magnesium ($Z=12$, config $2, 8, 2$)?

- (A) 1 (B) 2
(C) 6 (D) 8

Q.23 What is the chemical formula of Calcium Chloride (Calcium valency 2, Chlorine valency 1)?

- (A) CaCl (B) CaCl_2
(C) Ca_2Cl (D) Ca_2Cl_3

Q.24 Why do isotopes of an element share identical chemical properties?

- (A) Because they have the same mass number (B) Because they have the same number of valence electrons and identical electronic configurations
(C) Because they have the same number of neutrons (D) Because they are radioactive

Q.25 The total number of nucleons in an atom of Carbon-14 ($^{14}_6\text{C}$) is:

- (A) 6 (B) 8
(C) 14 (D) 20

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(A)	J.J. Thomson identified cathode rays as streams of negatively charged subatomic particles called electrons.
Q.2	(A)	Eugen Goldstein discovered canal rays (anode rays) consisting of positive ions.
Q.3	(A)	James Chadwick proved the existence of neutral neutrons by bombarding beryllium with alpha particles.
Q.4	(B)	Massive deflections of alpha particles led Rutherford to discover the tiny, dense, positively charged nucleus.
Q.5	(B)	An alpha particle consists of two protons and two neutrons, identical to a Helium-4 nucleus (He^{2+}), mass 4 u, charge $+2e$.
Q.6	(B)	Capacity $= 2n^2 = 2(2)^2 = 2 \times 4 = 8$ electrons.
Q.7	(B)	According to the Bohr-Bury octet rule, the outermost shell cannot hold more than 8 electrons.
Q.8	(A)	K-shell holds 2, L-shell holds 8, and M-shell holds the 1 remaining electron (2, 8, 1).
Q.9	(B)	Chlorine gains 1 electron to complete its octet: $\text{Valency} = 8 - 7 = 1$.
Q.10	(B)	Argon has a completely filled octet in its valence shell, giving it zero chemical combining capacity (valency 0).
Q.11	(B)	Mass number $A = Z + N$ (total number of protons and neutrons in the nucleus).
Q.12	(B)	Isotopes have the same number of protons but different numbers of neutrons (e.g., ^{12}C and ^{14}C).
Q.13	(B)	Isobars share the same mass number A (e.g., $^{40}_{18}\text{Ar}$ and $^{40}_{20}\text{Ca}$).
Q.14	(B)	Natural chlorine contains ^{35}Cl (75%) and ^{37}Cl (25%), yielding weighted average $\frac{35 \times 3 + 37 \times 1}{4} = 35.5$ u.
Q.15	(B)	Uranium-235 undergoes controlled nuclear fission in reactors, generating electricity.
Q.16	(A)	Cobalt-60 emits high-energy gamma radiation used to destroy malignant tumor cells.
Q.17	(A)	Iodine-131 is selectively concentrated by the thyroid gland, treating thyroid diseases.
Q.18	(B)	Protium (^1_1H) has $Z=1$ and $A=1$, having $1-1=0$ neutrons.
Q.19	(B)	An electron has a rest mass of 9.1×10^{-31} kg, which is about $1/1836$ (or $1/1840$) of a proton's mass.
Q.20	(C)	Neutrons are electrically neutral subatomic particles with zero net charge.
Q.21	(A)	Niels Bohr postulated stable stationary quantum orbits in 1913 to resolve atomic stability.
Q.22	(B)	Magnesium readily donates its 2 valence electrons from the M-shell to achieve an octet, giving valency 2.
Q.23	(B)	Criss-crossing valencies (Ca^{2+} and Cl^{-1}) yields CaCl_2 .
Q.24	(B)	Chemical reactivity depends on valence electronic structure, which is identical across isotopes.
Q.25	(C)	Nucleons are protons plus neutrons, which equals the mass number $A = 14$.